OSPE DAY 2

SPOT NO. 1.

1. Name the salivary glands labeled A, B, & C and the enzymes produced by them
   Ans: A → Parotid; (secretes salivary amylase); B → Submandibular; C → Sublingual (secretes salivary lipase)
2. Write down any 3 defense mechanism of saliva?
   Ans: IgA, lactoferrin, lysozyme, Proline rich proteins
3. How much saliva is produced per day?
   Ans: 1.5 L

SPOT NO. 2.

1. Define swallowing.
   Ans: It is the process of moving food from the mouth, by way of the pharynx and esophagus, into the stomach.
2. What are the phases of swallowing?
   Ans: Oral, Pharyngeal, Esophageal
3. Which sensory nerves initiate the swallowing reflex?
   Ans: 9^{th} and 10^{th}

SPOT NO. 3.

1. Name the type of electrical activity shown in A & B.
   Ans: A → Action Potential or Spike Potential; B → Slow wave potential (BER)
2. Name 3 stimuli that increase the number of electrical activity in A.
   Ans: Stretch, Acetylcholine, Parasympathetics
3. Which cells produce current B?
   Ans: Cells of Cajal

SPOT NO. 4.

1. Name the secretions released by cells labeled B and D.
   Ans: B cells → Intrinsic factor, HCl; D cells → Pepsinogen, Gastric Lipase, Rennin
2. Name any two stimuli that inhibit secretion by cell labelled F?
   Ans: Gastrin release is inhibited by: the presence of acid (primarily the secreted HCl) in the stomach (a case of negative feedback); somatostatin; secretin; GIP (gastric inhibitory peptide); VIP (vasoactive intestinal peptide); glucagon and calcitonin.
3. What is the function of the protein R-Binder?
   Ans: It binds with acid sensitive Vitamin B_{12} while it moves through the stomach.

SPOT NO. 5.

1. Identify A, B, C, and D?
   Ans: A → Maltose; B → alpha-limit Dextrins; C → Fructose; D → Lactose
2. Write the function of given enzymes present in brush border epithelium?
   Ans: Sucrase breaks down sucrose into glucose & fructose; **Isomaltase** acts on branching points releasing maltotriose, maltose and glucose; **Maltase** breaks down maltose into two glucose molecules; **Lactase** breaks down lactose into glucose and lactose
3. Why abdominal pain is produced in lactose intolerance?
   Ans: Undigested lactose is acted upon by colonic bacteria which produce gas that cause bloating and abdominal pain.